

LAB Task 5

SEMESTER: SPRING 2025

SUBMITTED BY:

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ROLL NO:

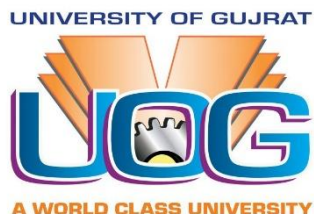
22011556-022

COURSE: IT-382 (SYSTEM AND NETWORK ADMINISTRATION)

PROGRAM: BS-INFORMATION TECHNOLOGY (SECTION A)

SUBMITTED TO:

MISS HIRA



Practical Implementation of Active Directory on Windows Server

1. Setting a Static IP Address Using Command Prompt (CMD)

Open CMD and list interfaces:

```
netsh interface ipv4 show interfaces
```

Then assign a static IP:

```
netsh interface ip set address name="Ethernet" static 192.168.1.10  
255.255.255.0 192.168.1.1  
netsh interface ip set dns name="Ethernet" static 192.168.1.10
```

2. Renaming the Server via CMD

Check hostname:

```
hostname
```

Rename to DC01:

```
wmic computersystem where name="%computername%" call rename name="DC01"  
shutdown /r /t 0
```

2. Installing Active Directory Domain Services (AD DS)

Open PowerShell as Administrator:

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
```

3. Promoting the Server to Domain Controller

Import-Module ADDSDeployment

Install AD DS Forest:

```

Install-ADDSForest `
-DomainName "tasmia.local" `
-DomainNetbiosName "TASMIA" `
-ForestMode "7" `
-DomainMode "7" `
-InstallDNS `
-SysvolPath "C:\Windows\SYSVOL" `
-LogPath "C:\Windows\NTDS" `
-DatabasePath "C:\Windows\NTDS" `
-SafeModeAdministratorPassword (ConvertTo-SecureString
"YourStrongPassword123" -AsPlainText -Force) `
-Force

```

4. Logging in After Domain Setup

Username: tasmia\Administrator

Password: <your password>

5. Creating Organizational Units (OUs), Users, and Groups

Create OUs:

```

New-ADOrganizationalUnit -Name "Students" -Path "DC=tasmia,DC=local"
New-ADOrganizationalUnit -Name "IT" -Path "DC=tasmia,DC=local"
New-ADOrganizationalUnit -Name "HR" -Path "DC=tasmia,DC=local"

```

Create a User:

```

New-ADUser -Name "Areeba Khan" -SamAccountName "areeba" -
UserPrincipalName "areeba@tasmia.local" `
-AccountPassword (ConvertTo-SecureString "AreebaPass123" -AsPlainText -
Force) `
-Path "OU=Students,DC=tasmia,DC=local" -Enabled $true -
PasswordNeverExpires $true

```

Create a Group:

```

New-ADGroup -Name "ITAdmins" -GroupScope Global -Path
"OU=IT,DC=tasmia,DC=local"
Add-ADGroupMember -Identity "ITAdmins" -Members areeba

```

6. Setting Up and Linking a Group Policy Object (GPO)

```

New-GPO -Name "BlockControlPanel"
New-GPLink -Name "BlockControlPanel" -Target "OU=IT,DC=tasmia,DC=local"

Set-GPRegistryValue -Name "BlockControlPanel" `

```

```
-Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer"  
`-ValueName "NoControlPanel" -Type DWord -Value 1
```

7. Forcing GPO Update on Client Machines

gpupdate /force

8. Joining a Client Computer to the Domain

```
netdom join %computername% /domain:tasmia.local  
/user:tasmia\Administrator /passwordd:*  
shutdown /r /t 0
```

9. Testing Domain Login

Username: tasmia\areeba

Password: AreebaPass123

10. Optional: File Sharing via CMD

```
mkdir C:\SharedFolder  
net share SharedFolder=C:\SharedFolder /grant:everyone,full  
Access: start \\DC01\SharedFolder
```

Active Directory setup using CMD and PowerShell completed successfully!

Included setup:

- Static IP ✓
- Server Renaming ✓
- AD DS Installation ✓
- Domain Promotion ✓
- OUs, Users, and Groups ✓
- GPO Setup ✓
- Domain Join ✓
- File Sharing (Optional) ✓